MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG1DSCHPY100 – BASICS OF PHYSIOLOGY

Duration : 2 Hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Identify the type of connective tissue connects muscles to bones. [K] [1]
   A) Ligament
   B) Tendon
   C) Cartilage
   D) Adipose tissue

2. Basic unit of life is ……. [K] [1]
   A) Cell
   B) Tissue
   C) Organ
   D) Molecule

3. The normal pH of blood is …….. [U] [2]
   A) 5.7
   B) 7.4
   C) 6.5
   D) 7.2

4. The accumulation of fluid in body is known as …….. [U] [2]
   A) Oedema
   B) Swelling
   C) Ascites
   D) Inflammation

5. …… is the colouring pigment in blood [K] [2]

A) Haemoglobin
B) Myoglobin
C) Protein
D) Albumin

6. The muscle present in heart ……. [U] [3]
   A) Sketelal muscle
   B) Cardiac muscle
   C) Smooth muscle
   D) Unstriated muscle

(1 x 6 = 6 marks)

Part B

Short Answer Type Questions
Answer **2** Questions
Each question carries **2** marks

7. Describe the structure and function of adherens junctions. [U] [1]
8. Discuss oedema. [U] [2]
9. Explain spleen. [U] [2]
10. Enumerate different types of muscles. [K] [3]

(2 x 2 = 4 marks)

Part C

Short Essay Type Questions
Answer **4** Questions
Each question carries **5** marks

11. Discuss homeostasis. [U] [1]
12. Classify the main types of connective tissue. Briefly describe their characteristics and functions. [U] [1]
13. Enumerate the functions of lymph. [K] [2]
14. Explain the main functions of the skeletal system and its importance to overall health. [U] [3]
15. Explain the properties of muscles. [K] [13
16. Discuss the types of bone cells. [U] [3]

(5 x 4 = 20 marks)

Part D

Long Essay Type Questions
Answer **2** Questions
Each question carries **10** marks

17. Explain structure of cell with a neat diagram. [U] [1]
18. Explain the composition of blood. [U] [2]
19. Explain the structure and functions of muscles. [U] [3]
20. Discuss how autocrine signalling contributes to cancer progression. [U] [1]

(10 x 2 = 20 marks)